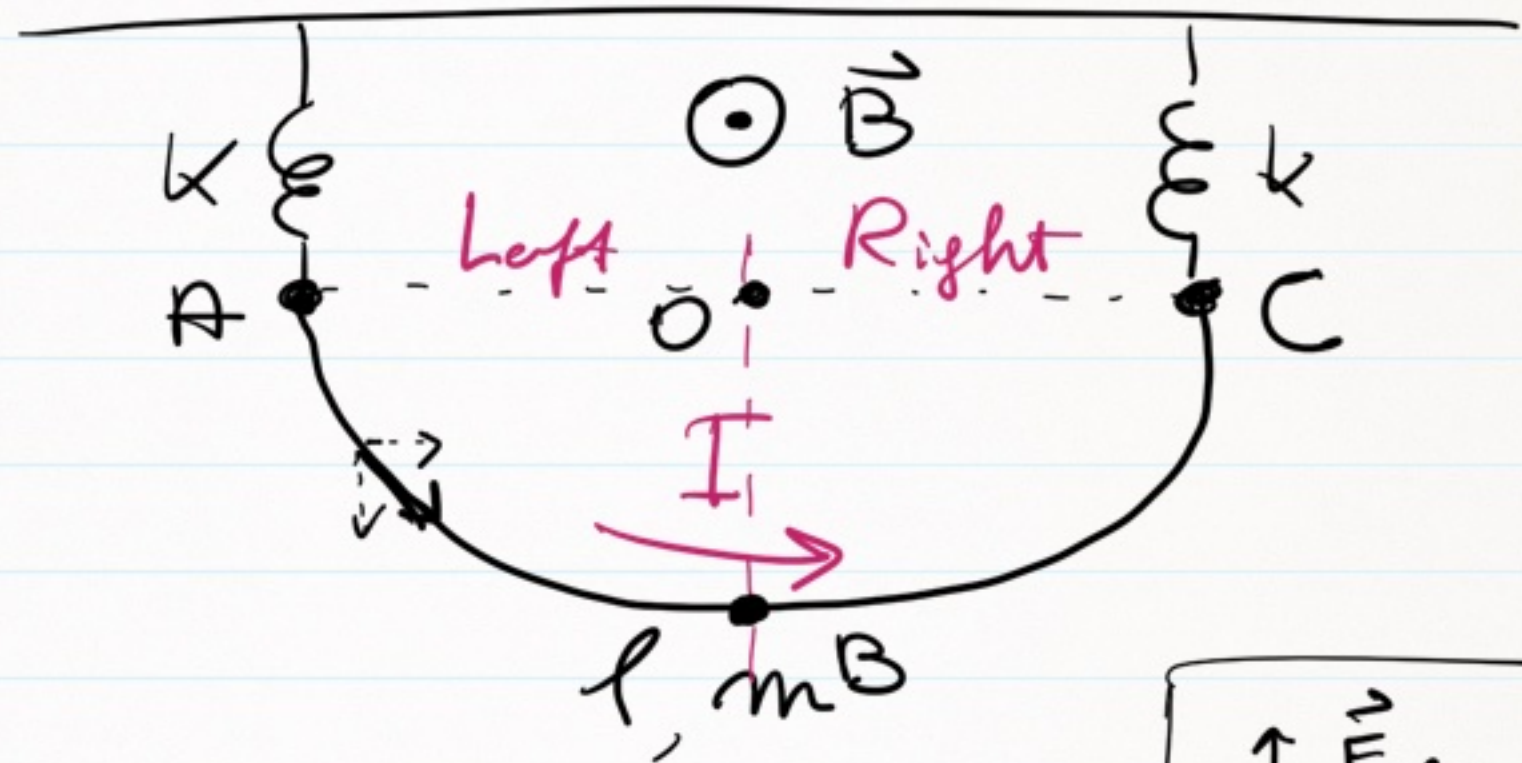
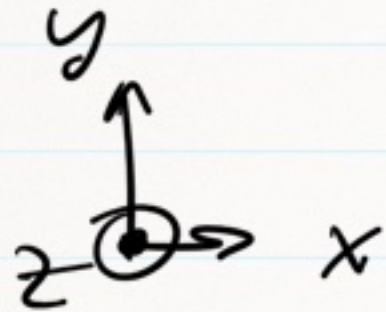


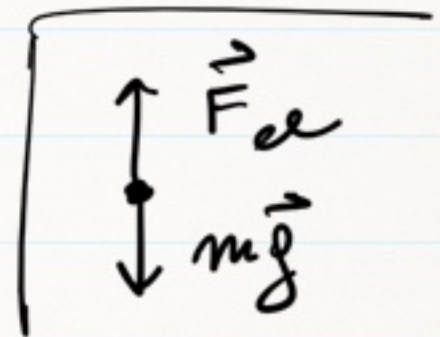
25/11/20

Ex. 1



(a) $\boxed{mg = 2k \cdot \Delta y_0}$

$$\Delta y_0 = \frac{mg}{2k} = \frac{50 \cdot 10^{-3} \cdot 9.8}{2 \cdot 10} = 2.45 \text{ cm}$$



(b) 2^a legge di Laplace

$$\boxed{d\vec{F} = I d\vec{\ell} \times \vec{B}}$$

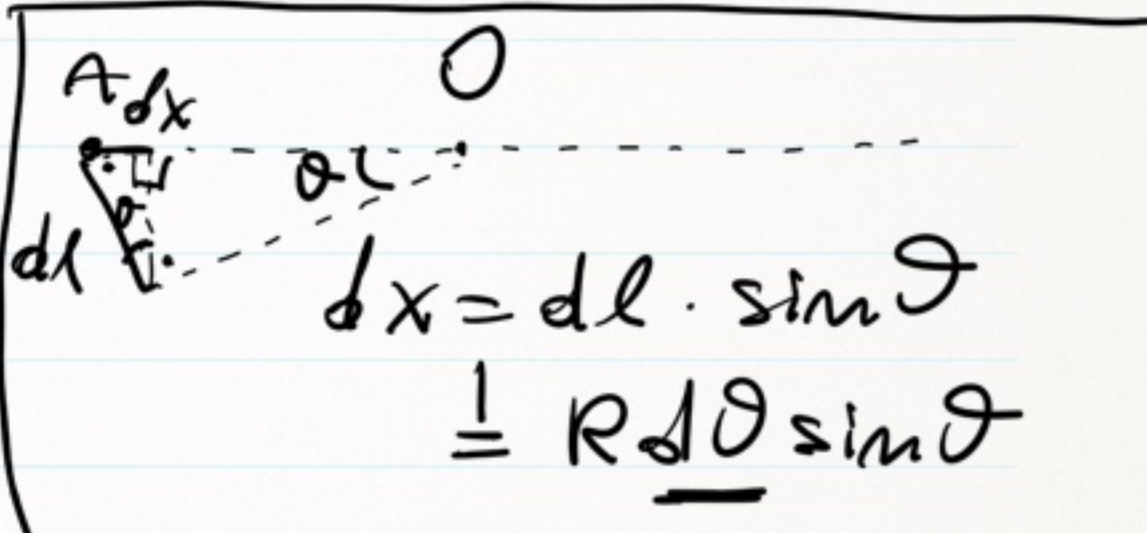
$$d\vec{\ell}_L = dx \hat{i} - dy \hat{j} \quad | \quad d\vec{\ell}_R = dx \hat{i} + dy \hat{j}$$

$$d\vec{F}_L = I (dx \hat{i} - dy \hat{j}) \times B \hat{k} =$$

$$= -I dx B \hat{j} + I dy B \hat{i}$$

$$d\vec{F}_R = -I dx B \hat{j} - I dy B \hat{i}$$

$$\begin{aligned} \vec{F} &= -2IB \int_{\pi/2}^0 dx \hat{j} \\ &= -2IBR \int_0^{\pi/2} \sin \theta d\theta \hat{j} \\ &= 2IBR [\cos \theta]_0^{\pi/2} \hat{j} = \end{aligned}$$



$dx = dl \cdot \sin \theta$
 $= R d\theta \sin \theta$

$$(l = \pi R) \quad \vec{F} = - \frac{2IBl}{\pi} \hat{j}$$

$$2K \cdot \Delta y_I = \frac{2IBl}{\pi}$$

$$\Rightarrow \Delta y_I = \frac{IBl}{K\pi} = \frac{2 \cdot 0.44 \cdot 0.628}{10 \cdot \pi} =$$

$$= 1.76 \text{ cm}$$

$$\Delta y_{\text{tot}} = \Delta y_{\theta} + \Delta y_I =$$

$$= 2.45 \text{ cm} + 1.76 \text{ cm} = 4.21 \text{ cm}$$

(c) corrente $\curvearrowright I$, altro componente
- $dx \hat{i}$ in $d\vec{\ell}$

$$\vec{F} = \frac{2 I B \ell}{\pi} \cdot \hat{j} \quad \text{compressione nulla}$$

$$\Delta y_I \cdot \hat{j} = - \frac{I B \ell}{K \pi} \hat{j}$$

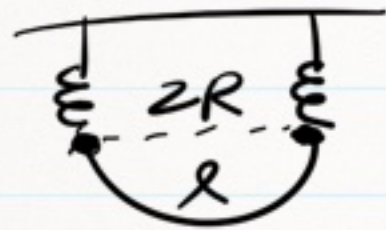
$$\Delta y_{\text{TOT}} = \Delta y_0 - \Delta y_I = 0.69 \text{ cm}$$

(d) corrente $\curvearrowright I^*$ t.c. $|\Delta y_{I^*}| = |\Delta y_0|$

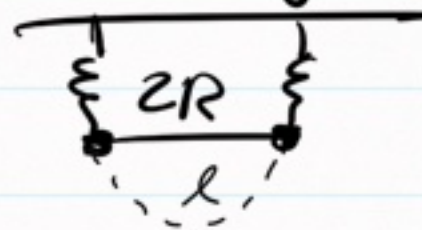
$$\frac{I^* B \ell}{\pi K} = \Delta y_0 \Rightarrow I^* = \frac{\Delta y_0 \pi K}{B \ell}$$

$$I^* = \frac{2.45 \cdot 10^2 \cdot \pi \cdot 10}{0.44 \cdot 62.8 \cdot 10^2} = 2.78 \text{ A}$$

Osservazione: se sostituisco il cavo di lunghezza l (semicirconferenza) con un cavo di lunghezza $2R$ (dritto)



$\Rightarrow$

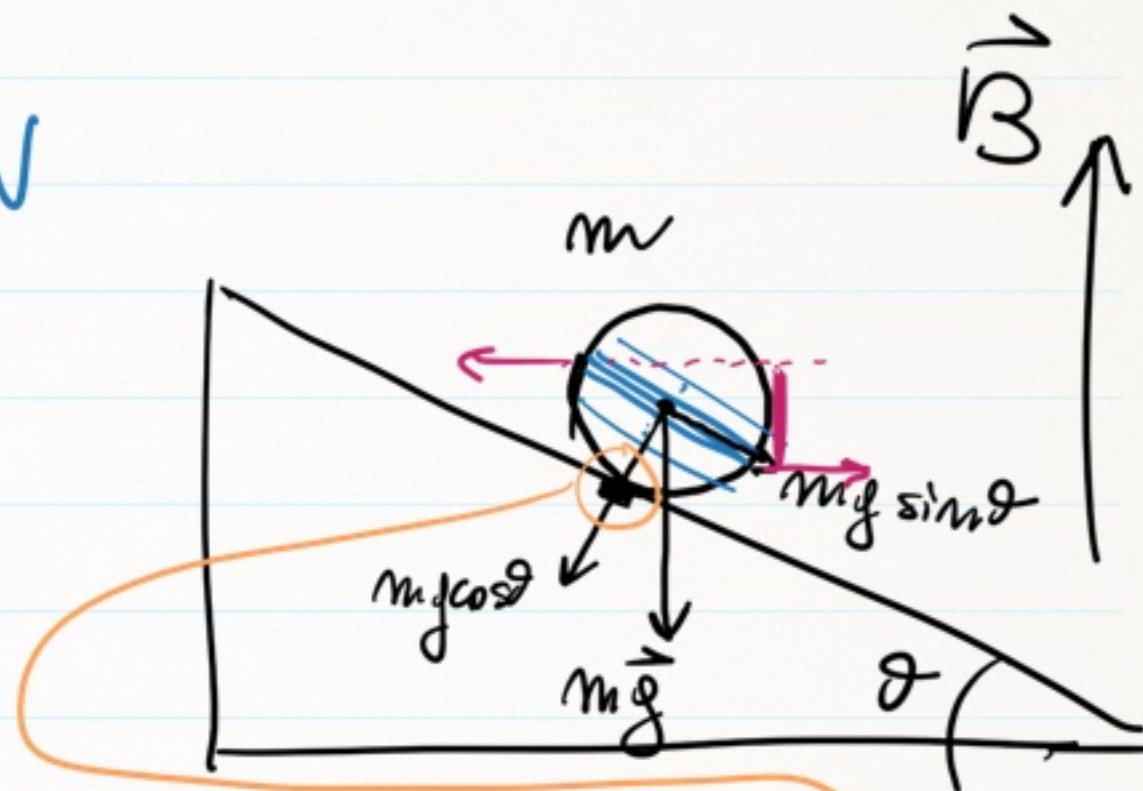
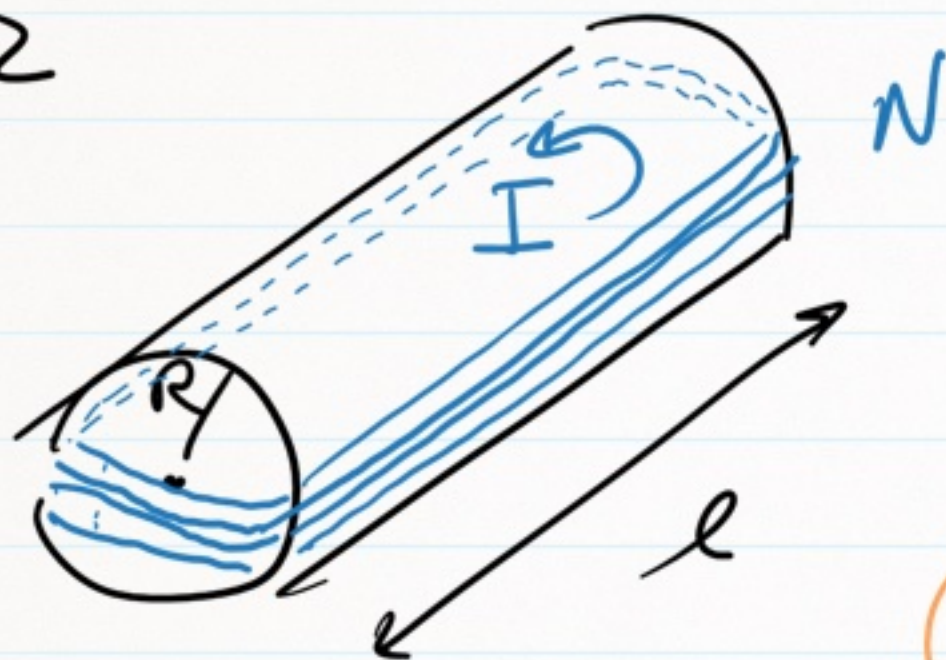


$$\vec{F} = -IB \int dx \hat{j} = -IB 2R \hat{j} = -\frac{2IBl}{\pi} \hat{j}$$

per cui la forza magnetica risulta come $\int_C d\vec{F}$, cioè dipende solo da PUNTO INIZIALE & FINALE!

stesso risultato!

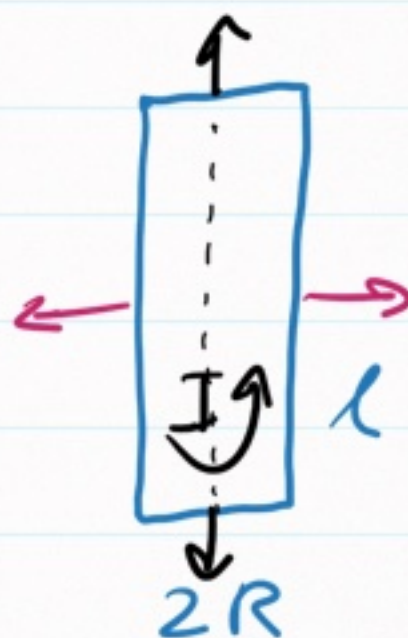
Ex 2

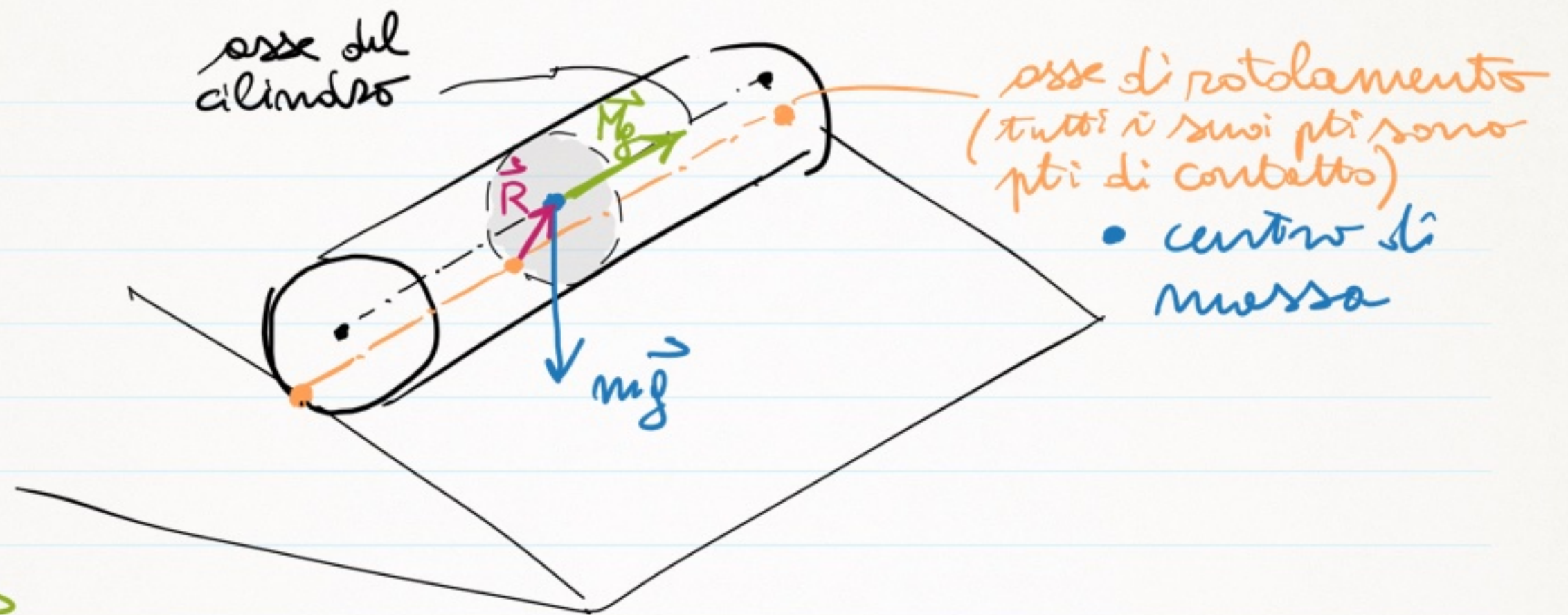


Momento forza peso: $\vec{M}_g = \vec{R} \times \vec{P}$ *per valutare M_g*

Momento dovuto a forza magnetica: ? *see next slide*

vista dall'alto: $\vec{B} \odot$



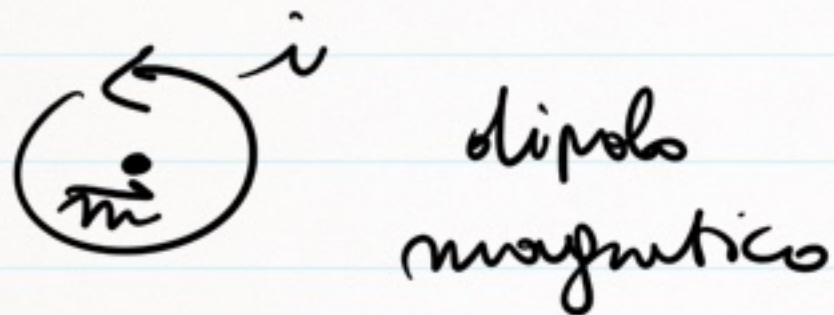


$\vec{M}_g$, momento della forza peso

$$\vec{M}_m = \vec{b} \times \vec{F}_m = F_m \cdot 2R \sin \vartheta$$

$$= N 2 I \ell B R \sin \vartheta$$

oppure secondo PRINCIPIO DI EQ. AMPÈRE



$$\vec{M}_m = \underline{\vec{m} \times \vec{B}}$$

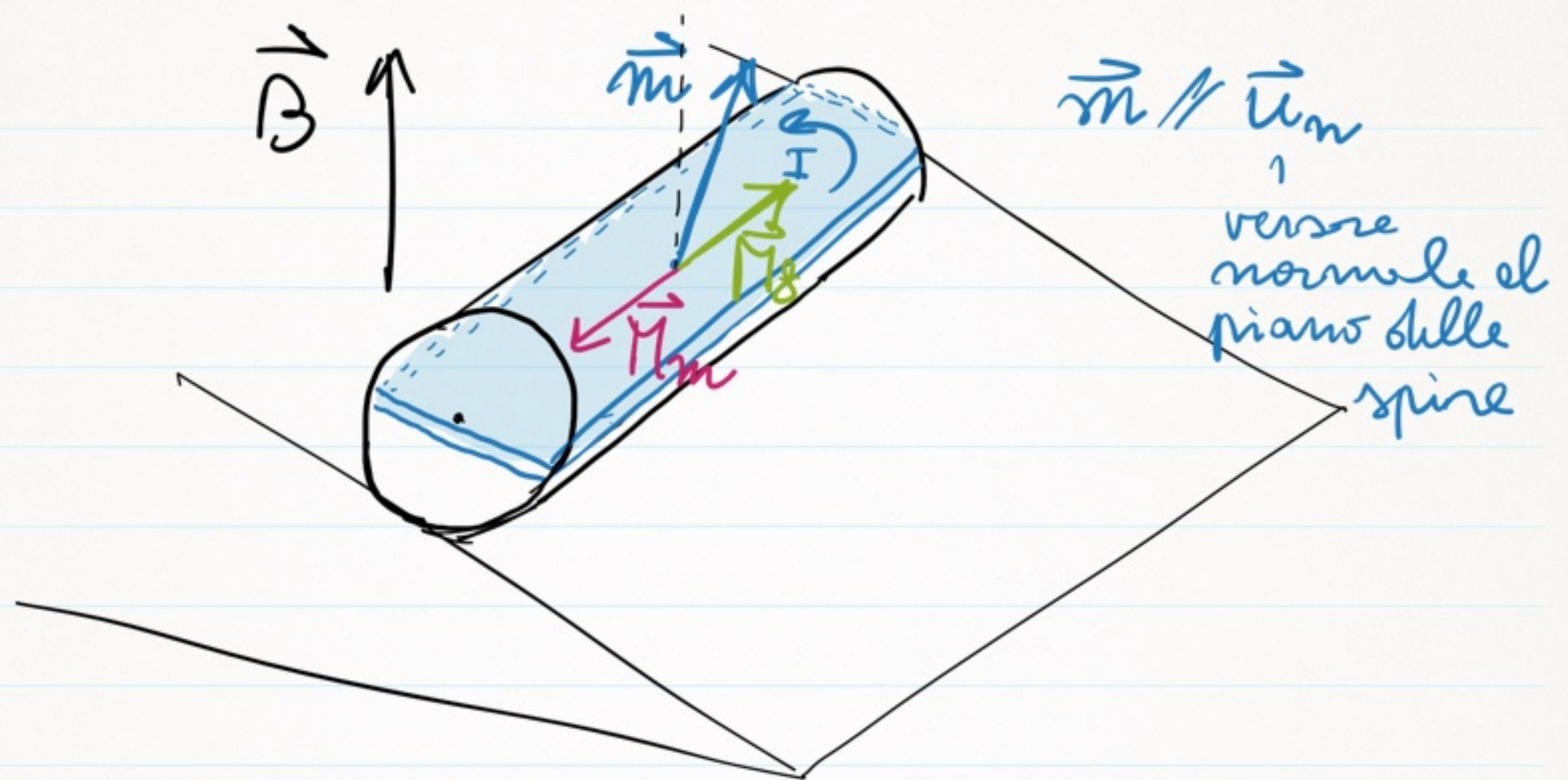
$$|\vec{m}| = I 2R \ell N$$

$$|\vec{M}_g| = |\vec{M}_m| \Rightarrow$$

$$I_{\min} = \frac{m g}{2 N \ell B}$$

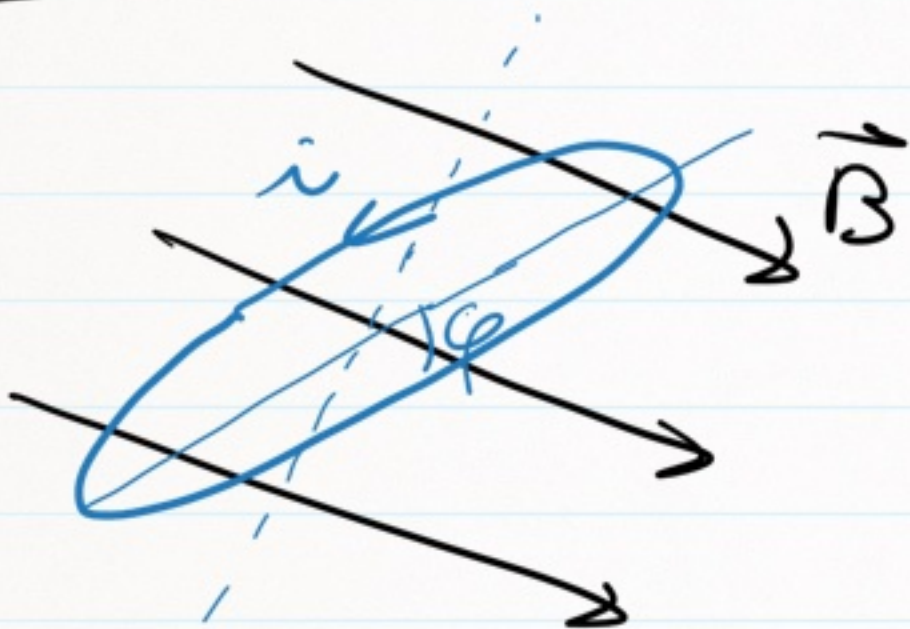
$$= 2.45 \text{ A}$$

$$(b) M_g = R (m g \sin \vartheta - \underline{\mu_s} m g \cos \vartheta)$$

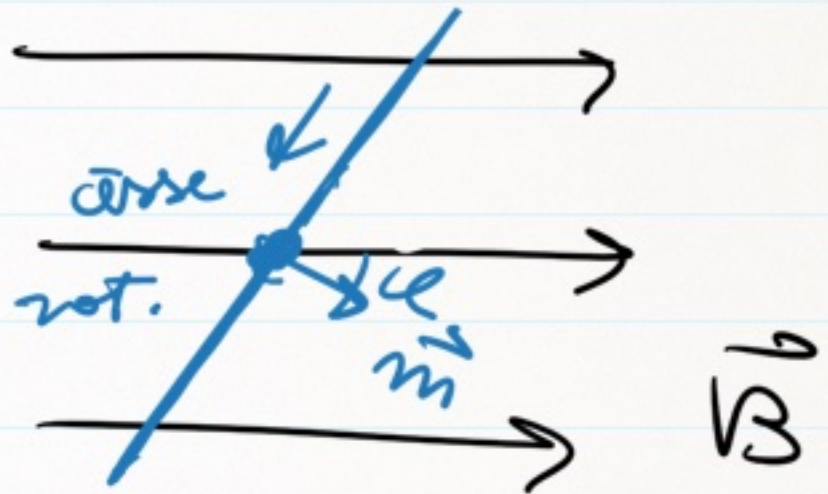


$\vec{M}_m$, momento della forza magnetica

EX. 3



vista di lato:



PRINC. EQUIV. AMPÈRE:

$$\vec{m} = i \sum \vec{u}_n \quad (\Sigma = \pi r^2)$$

$$\boxed{\begin{array}{l} \vec{M}_m \\ \vec{M} \\ \text{assiale} \end{array}} = \vec{m} \times \vec{B} = m B \sin \varphi$$
$$\frac{d\vec{L}}{dt} = I \cdot \frac{d\omega}{dt} = I \cdot \frac{d^2\varphi}{dt^2}$$

$$\boxed{I \frac{d^2\varphi}{dt^2} = -m B \sin \varphi} \Rightarrow$$

$$\sin \varphi \sim \varphi$$

$$\frac{d^2 \varphi}{dt^2} + \underbrace{\frac{m B}{I}}_{\omega^2} \varphi = 0 \quad \begin{array}{l} \text{OSC.} \\ \text{HARMONICO} \end{array}$$

$$\varphi(t) = \underline{\varphi_0} \cdot \sin(\omega t + \underline{\theta_0})$$

$$\frac{d\varphi}{dt} = \varphi_0 \omega \cos(\omega t + \theta_0) \quad \begin{array}{l} \text{partenza} \\ \text{de fermo} \end{array}$$

$$@t=0 \Rightarrow \varphi_0 \omega \cos(\theta_0) = 0$$

$$\theta_0 = \pi/2$$

$$\varphi_0 = 10^\circ$$

$$\varphi(t) = 10^\circ \cdot \cos(\omega t)$$

$$\omega = \sqrt{\frac{m B}{I}} = \sqrt{\frac{i \pi r^2 B}{m r^2}}$$

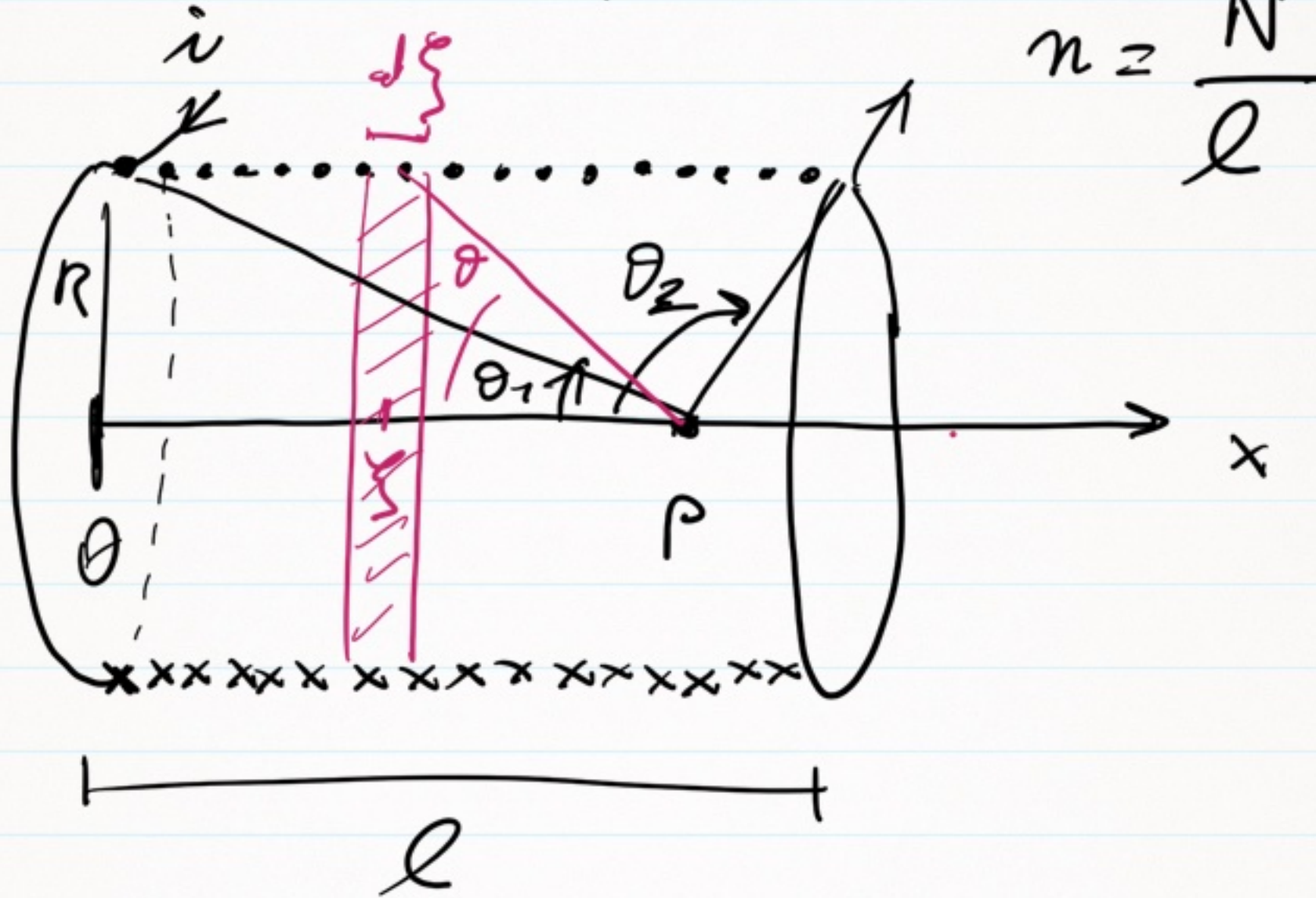
$$= \sqrt{\frac{3 \cdot 10^{-3} \pi \cancel{10^{-4}} 0,09}{10^{-2} \cancel{10^{-4}}}} =$$

$$= 0,29 \text{ rad/s}$$

$$T = \frac{2\pi}{\omega} = 23,45 \text{ s}$$

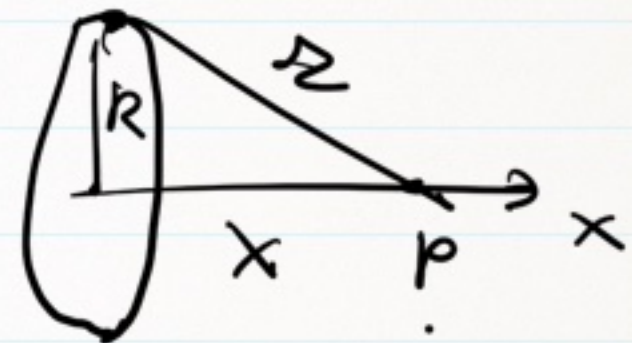
SOLENOIDE FINITO

Ex 7



Visto a lezione: 1 spira

$$\vec{B}(P, \text{asse spira}) = \frac{\mu_0 I R^2}{2 \lambda^3} \vec{u}_\lambda$$



$$\frac{d\vec{B}(P)}{d\xi} = n d\xi \cdot \frac{\mu_0 I R^2}{2 [R^2 + (x-\xi)^2]^{3/2}} \vec{u}_n$$

null'essex

$$\tan \theta = R / (x - \xi) \Rightarrow x - \xi = \frac{R}{\tan \theta}$$

$$-d\xi = -\frac{R d\theta}{\sin^2 \theta}$$

$$d\vec{B}(P) = \frac{n \mu_0 I R^2}{2 \left[R^2 + \left(\frac{R}{\tan \theta} \right)^2 \right]^{3/2}} \cdot \frac{R d\theta}{\sin^2 \theta} \vec{u}_n$$

$$d\vec{B}(P) = \frac{\mu_0 I \sin\theta d\theta}{2}$$

$$\vec{B}(P) = \frac{\mu_0 I}{2} \int_{\theta_1}^{\theta_2} \sin\theta d\theta \vec{u}_n =$$

$$= \frac{\mu_0 I}{2} (\cos\theta_1 - \cos\theta_2) \vec{u}_n$$

$$\cos\theta_1 = \frac{x}{\sqrt{x^2 + R^2}}, \quad \cos\theta_2 = -\cos(\pi - \theta_2) =$$

$$= -\frac{l-x}{\sqrt{(l-x)^2 + R^2}}$$

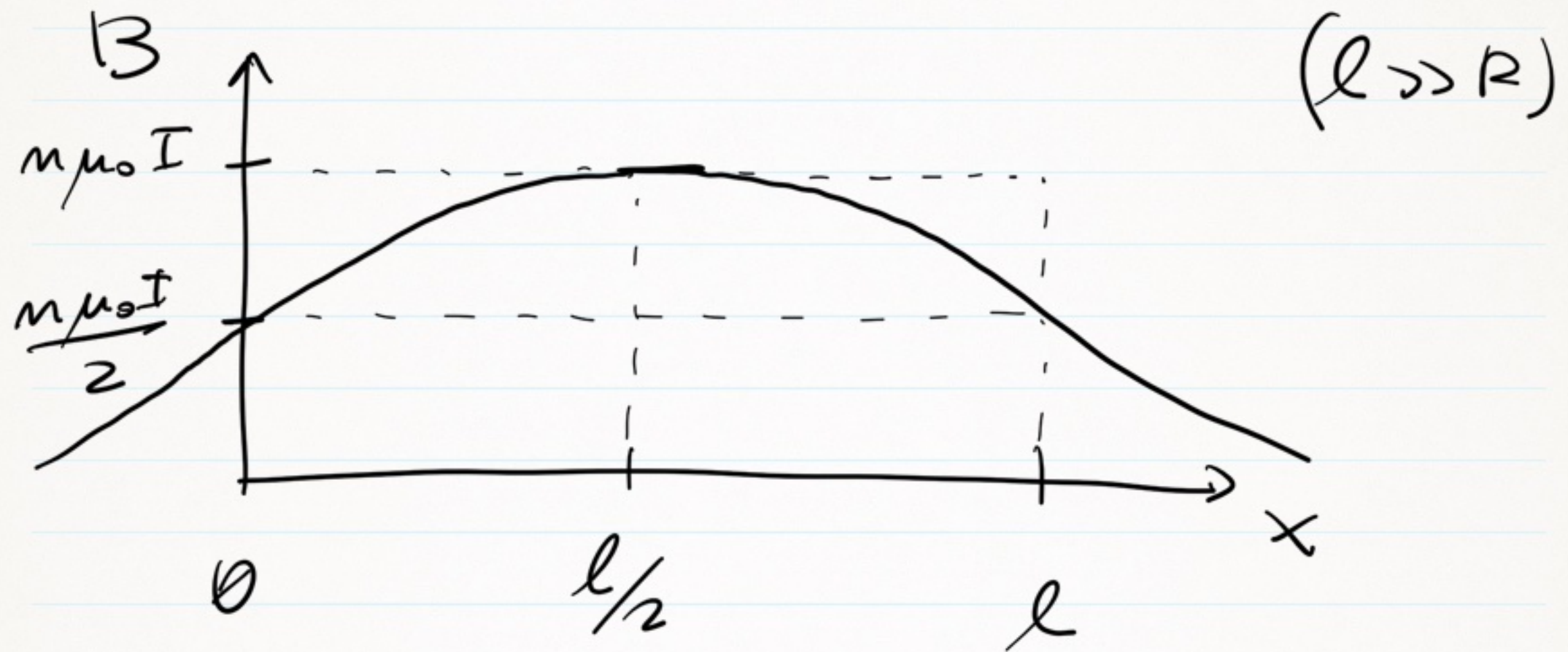
$$B(x) = \frac{n\mu_0 I}{2} \cdot \left(\frac{x}{\sqrt{x^2 + R^2}} + \frac{l-x}{\sqrt{(l-x)^2 + R^2}} \right)$$

$$B\left(x = \frac{l}{2}\right) = \frac{n\mu_0 I}{2} \cdot \frac{l}{\sqrt{\frac{l^2}{4} + R^2}}$$

$$l \gg R \rightarrow n\mu_0 I$$

$$\left[4\pi \cdot 10^{-7} \frac{\text{H}}{\text{m}} \right]$$

$$B(x=0) = \frac{n\mu_0 I}{2} \cdot \frac{l}{\sqrt{l^2 + R^2}} \xrightarrow{l \gg R} \frac{n\mu_0 I}{2}$$



Conclusione: il campo sull'asse al centro
 è pari a quello di un solenoide
 indefinito (e.g. @ centro il contributo
 delle spine è il massimo) MA
 dal centro verso le estremità
 il campo diminuisce...